

Front wing in ground effect beside a rotating wheel

Steady RANS setup and verification report

OpenFOAM v2606 (native) • `snappyHexMesh` + `simpleFoam` • case S1.42_h0.13_AOA8_W0.63

1 Case at a glance

Configuration		Mesh	
wing half-span	$S/c = n/a$	cells	n/a
ride height	$h/c = n/a$	background cells	n/a
angle of attack	n/a°	layer coverage	92.7 %
wheel width	$W/c = n/a$	<code>checkMesh</code> clean	no
Flow		Cost, 20 MPI ranks	
freestream	n/a m/s	meshing	34 min
chord	n/a m	solving	n/a min
Re_c	n/a	per iteration	8.62 s
ρ	n/a kg/m ³	iterations	70
ν	n/a m ² /s	converged	no

Headline result. Force *coefficient-areas*, mean over the last 50 iterations. $A_{\text{ref}} = 1 \text{ m}^2$ on purpose, so these are $C_L \cdot A$ and $C_D \cdot A$ in m² — the quantity the Diasinos papers tabulate. Negative lift is downforce.

	$C_L \cdot A$ [m ²]	$C_D \cdot A$ [m ²]	spread of $C_L \cdot A$
wing	n/a	n/a	n/a %
wheel	n/a	n/a	n/a %
total	n/a	n/a	n/a %

Mean y^+ : wing n/a, wheel n/a, maximum n/a.

2 Software and reproducibility

- **OpenFOAM v2606** (ESI/OpenCFD), built from source. Native only — no commercial solver, no in-house libraries.
- Mesh `snappyHexMesh`, solve `simpleFoam` (segregated, SIMPLEC), turbulence `kOmegaSST`.
- Geometry generated parametrically from the Diasinos specification; 420 configurations available.
- Everything else is `python3` standard library. No `numpy/pandas` needed to run a case.

To reproduce this case.

```
cp env/project.env.example env/project.env # set OPENFOAM_BASHRC and NP
cd Diasinos_Geometry_Generator && .venv/bin/python -m geometry_generator.generate
cd ../RANS_Simulations
python3 prepare_case.py --span 1.42 --height 0.13 --aoa 8 --width 0.63
cd S1.42_h0.13_AOA8_W0.63 && ./Allrun
```

3 Geometry

- Coordinate system: x streamwise (wheel centre at $x = 0$), y spanwise outboard, z up with the ground at $z = 0$. Half model, symmetry at $y = 0$.
- Chord $c = 75$ mm. Inverted NACA 4412, blunt trailing edge 0.5 mm. Wheel diameter $1.17c$, shoulder radius $0.067c$.
- **The generator writes millimetres**; everything in the simulation is metres. **Allmesh** scales by 0.001 on the way in.
- The *plinth* is a collar that deliberately pierces the ground plane (down to $z = -6$ mm) so the contact patch is a clean cut instead of a tangential singularity.

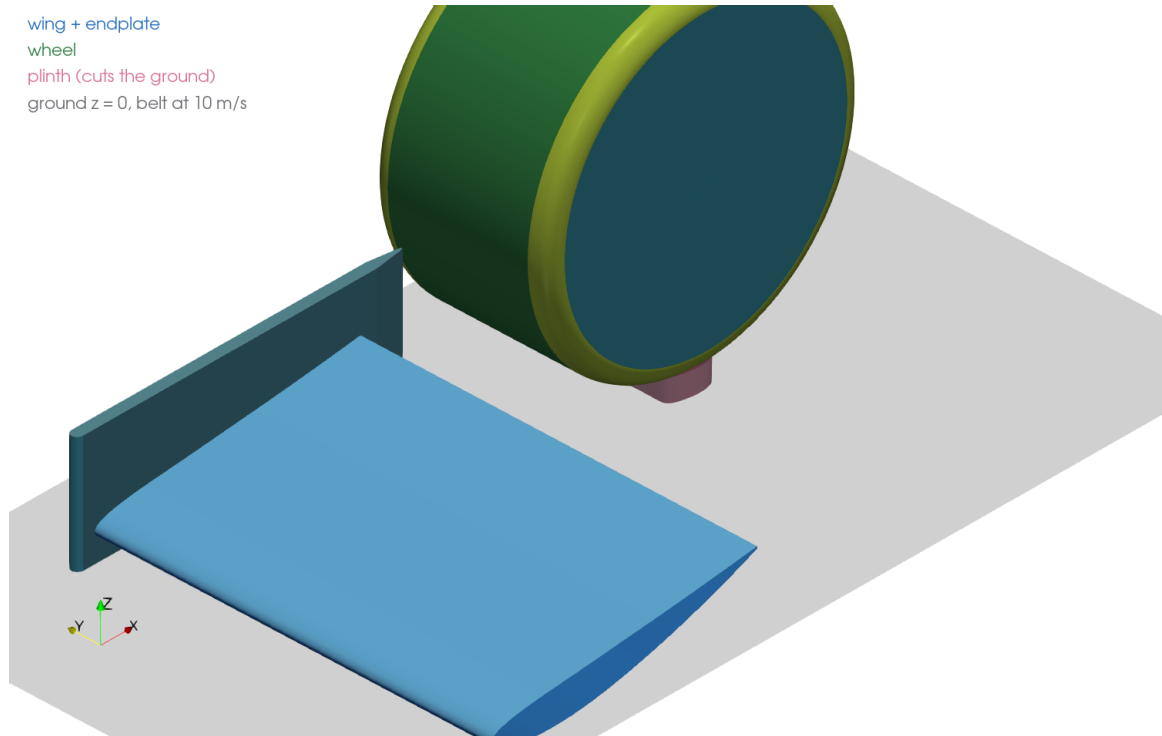


Figure 1: Geometry and the ground plane. The wing sits in ground effect ahead of and inboard of the wheel; the plinth passes through the ground.

Known geometry limitations.

- The wing is **prismatic**, extruded between two spanwise stations. Geometrically exact for a constant section; it does mean the triangles have a very high aspect ratio, which **surfaceCheck** reports as poor quality. Not an error.
- **Two duplicate triangles** on the endplate outer edge — a defect in the generator. Two facets out of 1142; **snappyHexMesh** tolerates them. Reported by **Allmesh** and deliberately non-fatal.
- Both STLs consist of two disjoint parts (wing + endplate, wheel + plinth), so **surfaceCheck** says “not closed”. Expected: no **mode inside** refinement region is attached to either body.
- Wheel tessellation was raised from 180 to 900 circumferential segments, so the facet is 0.31 mm against the 0.293 mm cell. At 180 segments the facets were 1.53 mm — five times coarser than the cell, i.e. the mesh resolved the wheel as a 180-gon with a visibly wavy surface. *This did not improve the layer coverage on the tread (63.4% before and after)*; it was done for geometric fidelity alone.

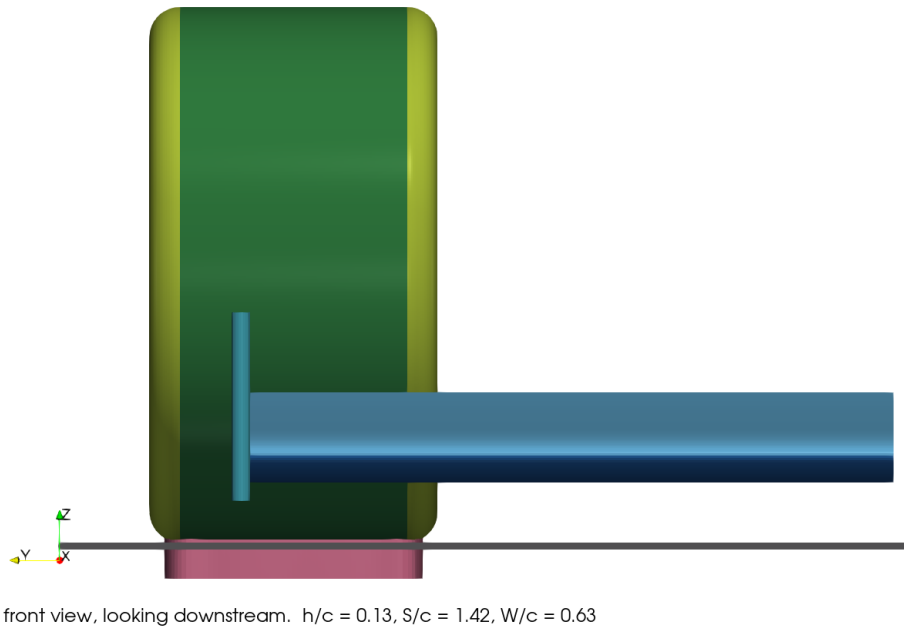


Figure 2: Front view, looking downstream. The endplate (light blue) falls inside the wheel silhouette: at $S/c = n/a$ and $W/c = n/a$ the spanwise gap is $-0.45c$, i.e. wing and wheel overlap.

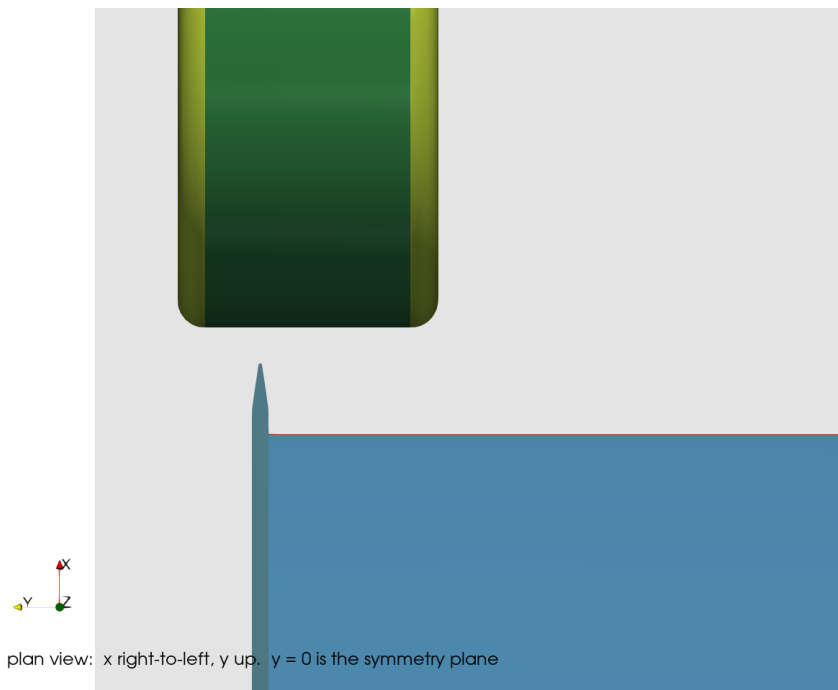


Figure 3: Plan view. $y = 0$ is the symmetry plane.

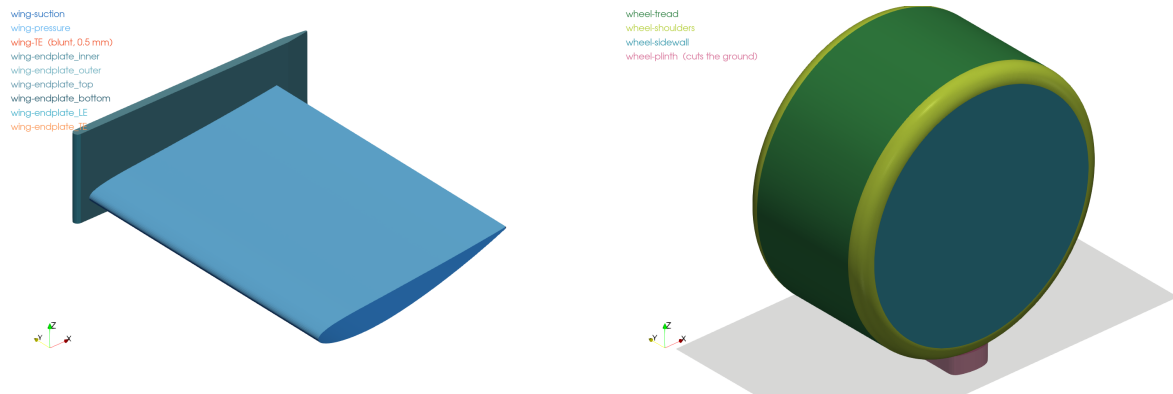


Figure 4: Named STL regions. `snappyHexMeshDict` refers to these names directly to set refinement level and patch name, so renaming a region in the generator breaks the mesh setup.

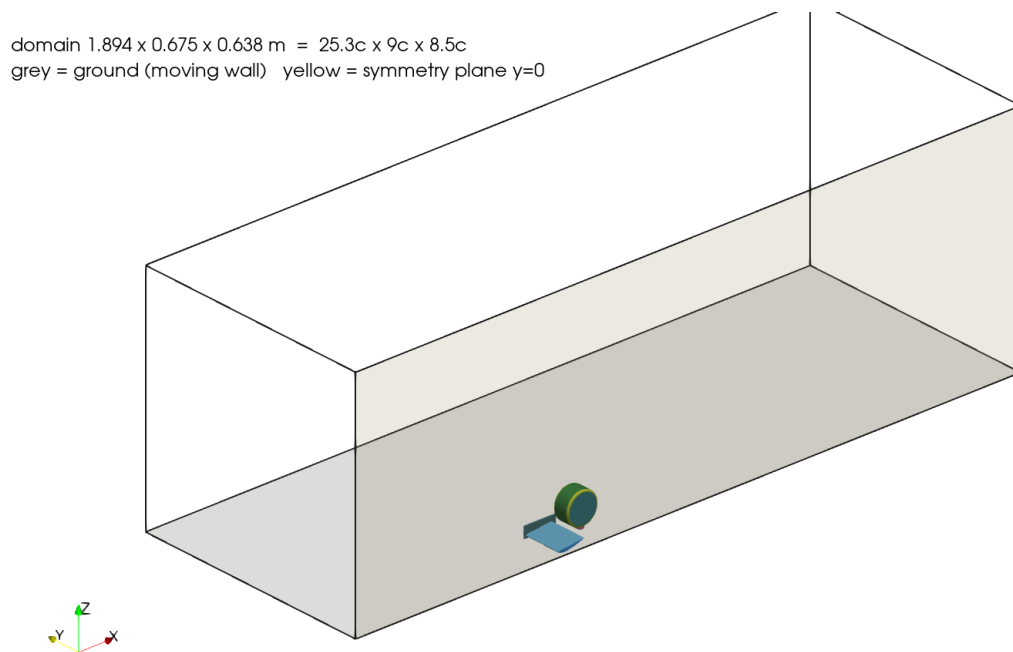


Figure 5: Computational domain, $1.894 \text{ m} \times 0.675 \text{ m} \times 0.638 \text{ m} = 25.3c \times 9c \times 8.5c$. Grey: moving ground. Yellow: symmetry plane.

4 Domain and boundary conditions

patch / group	type	U	p	k, ω, ν_t
inlet	patch	fixedValue (1000)	zeroGradient	turbulent inlet
outlet	patch	pressureInletOutletVelocity	fixedValue 0	inletOutlet
symmetry	symmetry	symmetry	symmetry	symmetry
side, sky	patch	slip	slip	slip / calculated
ground	wall	fixedValue (1000)	zeroGradient	wall functions
wingGroup	wall	noSlip	zeroGradient	wall functions
wheelRotating	wall	rotatingWallVelocity	zeroGradient	wall functions
wheelGround	wall	fixedValue (1000)	zeroGradient	wall functions

- **Rolling road:** the belt translates downstream at the freestream speed, so the ground is at rest relative to the far-field air.
- **Rotating wheel:** $\omega = -227.92$ rad/s about (0 1 0) through (0, 0, 0.043875 m). `rotatingWallVelocity` imposes $\omega \times r$ without moving the mesh — correct for steady RANS. The sign follows from requiring the contact point to move with the belt: $\omega_y = -U_\infty/R$. Reversing it spins the wheel backwards and still converges happily.
- The **wheel-plinth** patch is in **wheelGround**, not **wheelRotating**: it is the collar through the ground and moves with the belt.
- **side** and **sky** are **patch**, not **wall**, on purpose — a wall there would enter the **meshWave** wall-distance field that **kOmegaSST** blends on, and would appear in the force groups.

5 Mesh

- Background: $101 \times 36 \times 34$ **cubic** cells of 18.75 mm ($= 0.25c$), n/a cells. Cubic matters: each refinement level halves the edge, so the level-to-size table only holds from a cube.
- Level sizes: L4 1.17 mm, L5 0.586 mm, L6 0.293 mm, L7 0.146 mm.
- Per-region surface levels, distance refinement around both bodies (L5 within 15 mm, L4 within 30 mm), and nine **searchableBox** zones.
- Final mesh n/a cells.

refinement level	cells	share [%]
L0	110 876	0.6
L1	19 040	0.1
L2	519 518	2.6
L3	592 809	3.0
L4	4 255 806	21.4
L5	11 921 774	60.1
L6	1 118 543	5.6
L7	1 312 372	6.6

5.1 Prism layers

Four layers from 0.14 mm with expansion 1.3 (total 0.87 mm) on wing and wheel, three from 0.25 mm on the ground. This targets $y^+ \approx 2.5$, which is why the wall functions are the all- y^+ variants.

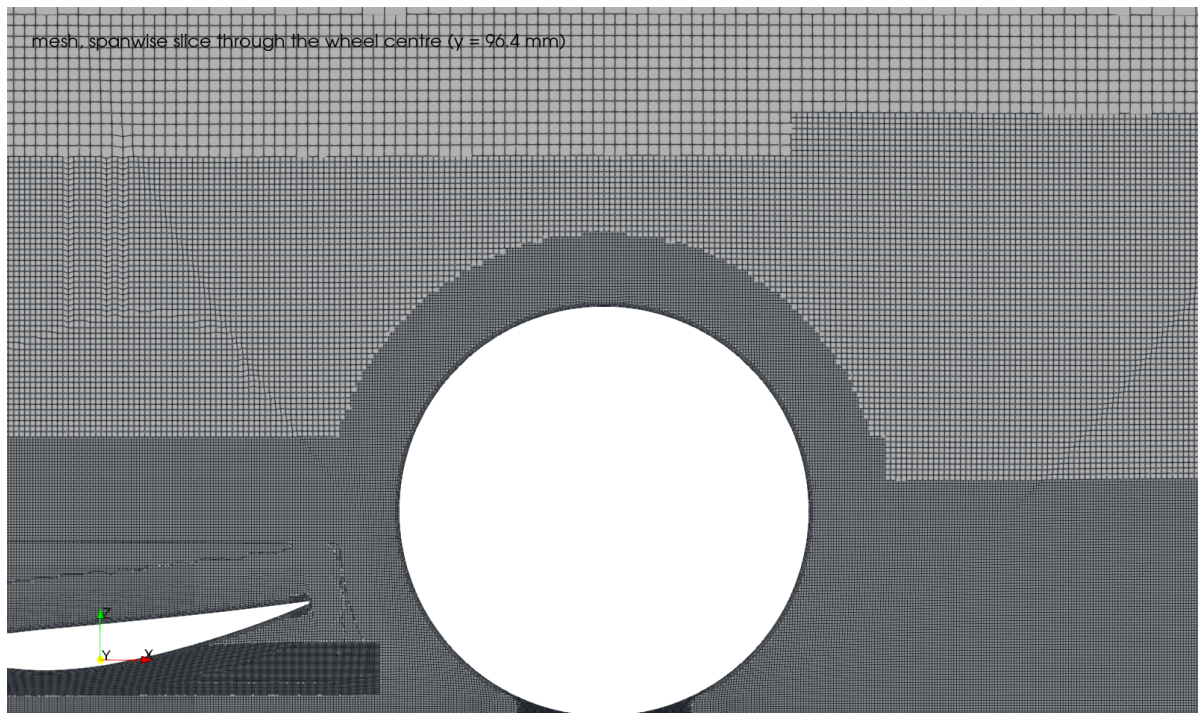


Figure 6: Mesh on a spanwise slice through the wheel centre. The nested refinement zones and the wake box are visible.

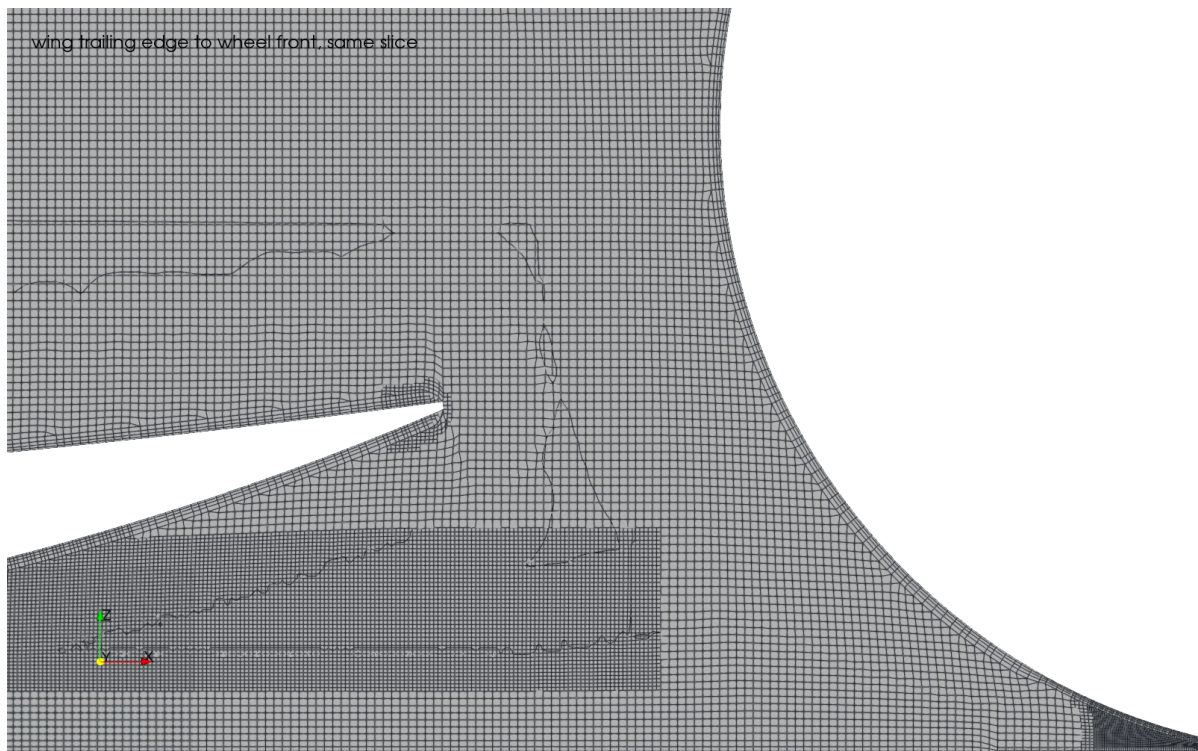


Figure 7: Same slice, wing trailing edge to wheel front — the region the whole study is about.

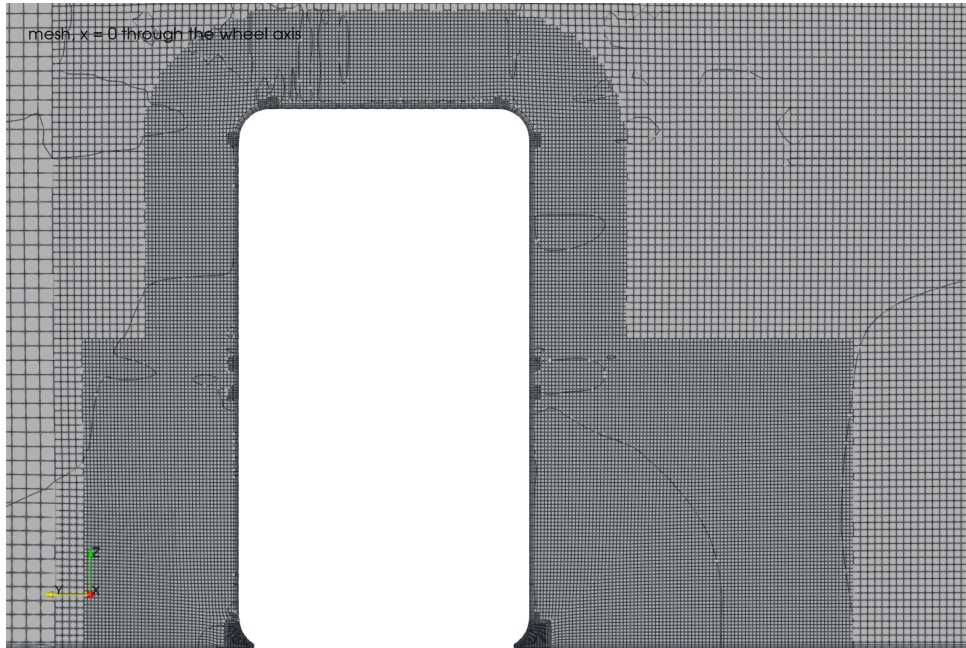


Figure 8: Slice through the wheel axis ($x = 0$). Looking downstream, so y increases to the left.

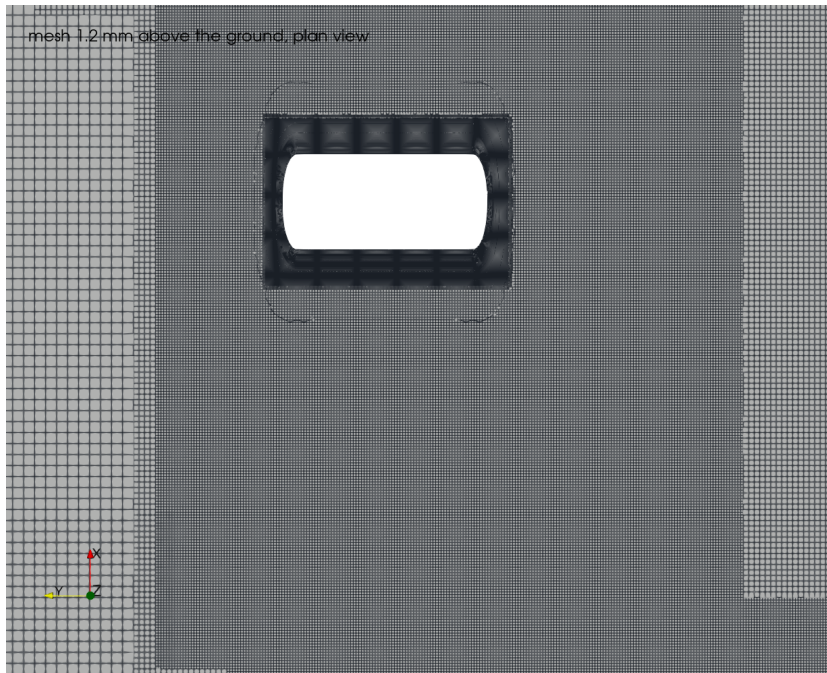


Figure 9: Mesh 1.2 mm above the belt. The contact-patch refinement box is the fine square under the wheel.

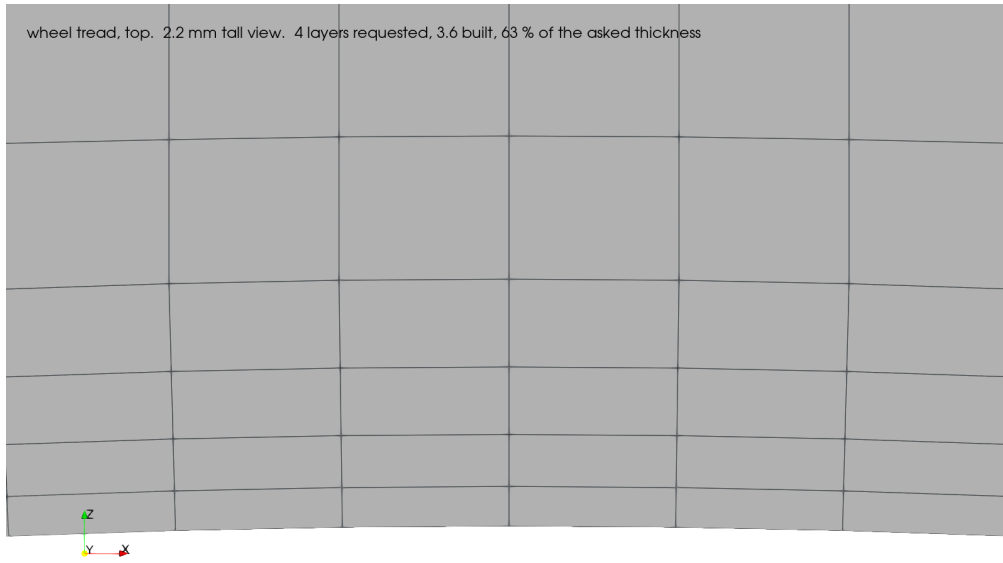


Figure 10: Wheel tread at the top. Prism layers on the 0.31 mm-tessellated tread.

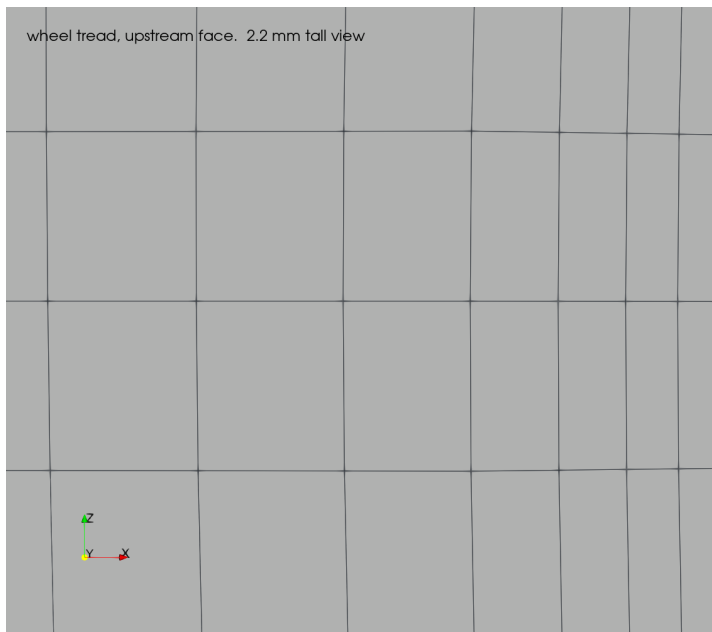


Figure 11: Wheel tread, upstream face.

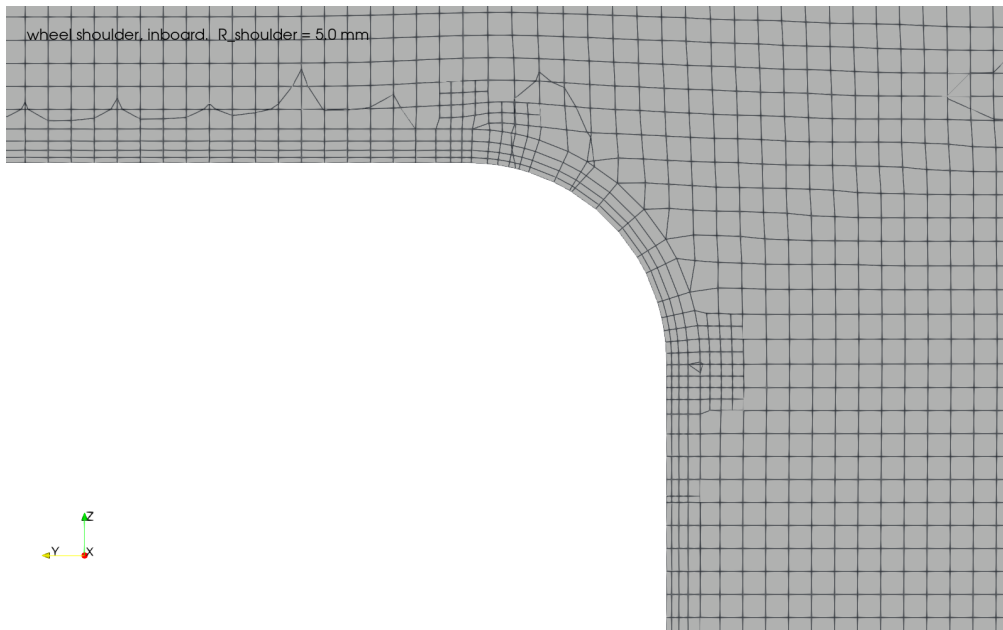


Figure 12: Wheel shoulder, inboard. The 5 mm shoulder radius is the tightest convex curvature on the wheel.

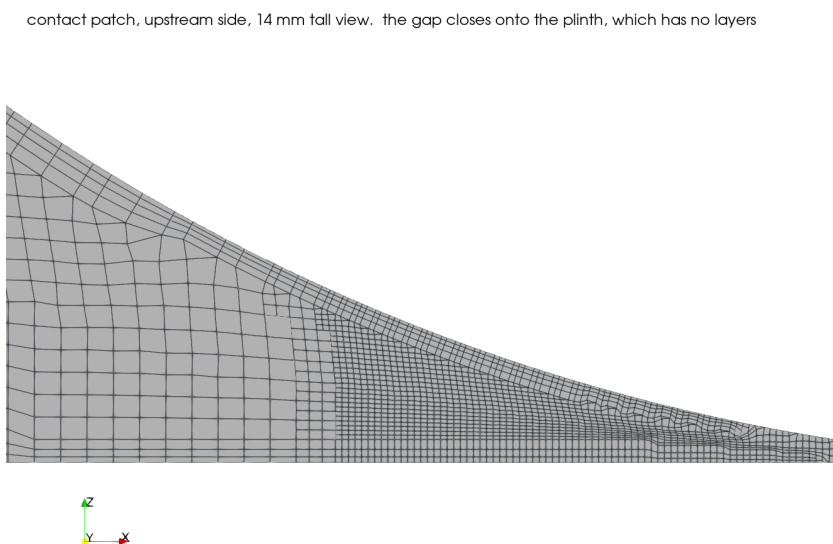


Figure 13: Contact patch. The plinth cuts the ground plane at $z = 0$; this is where the layers fight each other.

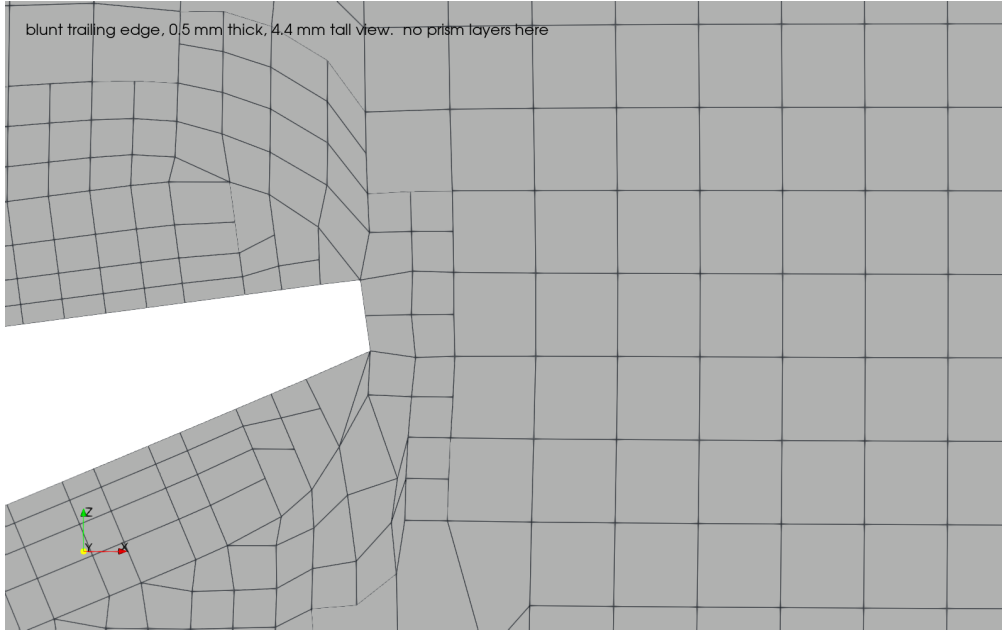


Figure 14: Blunt trailing edge, 0.5 mm thick. No prism layers — see below.

patch	faces	layers wanted	layers built	thickness [%]
ground	225301	3	2.92	82.3
wing-suction	31401	4	3.88	95
wing-pressure	27763	4	3.87	96.1
wing-TE	752	2	0	0
wing-endplate_inner	14340	4	3.47	79.4
wing-endplate_outer	16775	4	3.75	95.1
wing-endplate_top	3256	4	2.85	76
wing-endplate_bottom	3256	4	2.85	76.4
wing-endplate_LE	1392	4	3.87	97.4
wing-endplate_TE	212	2	0	0
wheel-tread	68829	4	3.59	63.4
wheel-shoulders	37577	4	3.08	67.3
wheel-sidewall	33128	4	3.99	99.9
wheel-plinth	2279	2	0	0

Reading the layer table.

- Overall coverage 92.7 %.
- The patches in **bold** got **no layers at all**. These are **wing-TE**, **wing-endplate_TE** and **wheel-plinth** — sharp convex edges and the plinth/ground intersection. Asking for fewer and thinner layers does not change it: the limit is topology, not thickness.
- **This is accepted.** **snappyHexMesh** is known for incomplete layer coverage at sharp edges and where two layered surfaces meet. All three patches sit in L7 cells (0.146 mm), so the near-wall spacing is comparable to the layered regions anyway.
- The number to judge the near-wall treatment by is y^+ (section 8), not the layer count.

5.2 Mesh quality

Quality limits are deliberately relaxed (**maxNonOrtho** 75, **maxInternalSkewness** 6, **minTetQuality** -10^{30}), carried over from the previous HELYX setup. Enforcing a positive tet quality with absolute layer thicknesses on curved surfaces costs most of the layer coverage.

checkMesh clean: **no** (0 findings).

- no failures reported

These are the expected consequence of the relaxed limits. The solver runs stably on this mesh; if it ever diverges, `meshQualityDict` is the first thing to question.

6 Physics and numerics

turbulence model	kOmegaSST
ν_t wall function	nutUSpaldingWallFunction (continuous all y^+)
k wall function	kLowReWallFunction
ω wall function	omegaWallFunction
momentum convection	bounded Gauss linearUpwindV grad(U)
turbulence convection	bounded Gauss upwind (first order, deliberate)
gradients	cellLimited Gauss linear 1
laplacian	Gauss linear corrected
wall distance	meshWave
pressure-velocity	SIMPLEC (consistent yes)
p solver	GAMG, tolerance 10^{-7} , relTol 0.01
U, k, ω	smoothSolver GaussSeidel
relaxation	U 0.9, k and ω 0.7
residualControl	p, U : 10^{-5} ; k, ω : 10^{-4}
initialisation	potentialFoam, divergence-free start

Turbulence convection is first order on purpose. Second-order k/ω transport is the usual reason residuals stall on a mesh of this kind, and the effect on the integrated forces is small. Momentum convection is second order.

Differences against the previous HELYX setup. Relevant whenever the two are compared:

- **No curvature correction** (HELYX used Menter–Smirnov). The endplate and wheel vortex cores will diffuse somewhat faster.
- **No separation fix** (HELYX used the Rumsey form). Separation onset on the suction side may differ.
- The adaptive wall treatment *is* reproduced, by the all- y^+ wall functions above.
- HELYX ran a coupled solver pseudo-transiently; this is segregated and steady, so iteration counts are not comparable.

7 Convergence

figure conv_residuals.png not generated

Figure 15: Initial residuals per SIMPLE iteration, with the `residualControl` thresholds marked.

figure conv_forces.png not generated

Figure 16: Force history. The grey band is the window averaged for the reported result.

figure conv_forces_zoom.png not generated

Figure 17: Settling of the total $C_L \cdot A$ over the final iterations.

How to judge this.

- residualControl satisfied: **no**.
- Final initial residuals: p n/a, U_x n/a, U_z n/a, k n/a, ω n/a.
- **The force spread matters more than the residual.** A converged residual with $C_L \cdot A$ still swinging by several percent means an unsteady flow being forced into a steady solve. Here the total spread over the averaging window is n/a %.
- Residual *fields* are written as well, so a stalled case can be opened in ParaView to see *where* it will not settle.

8 Results

figure res_cp_wing.png not generated

Figure 18: Pressure coefficient on the wing suction side.

figure res_cpt_slice.png not generated

Figure 19: Total pressure coefficient on streamwise slices — vortex cores and wake losses.

9 Assessment

What supports trusting these numbers.

- Background mesh exactly as designed (n/a cells), cubic, dividing the domain without remainder.
- Refinement reproduces the previous validated HELYX setup level for level, and the final cell count agrees with it closely.
- Patch groups resolve correctly — patchSummary confirms noSlip on the wing, rotatingWallVelocity on the wheel and the belt velocity on the plinth.
- Wheel rotation sign verified analytically against the belt speed.
- Divergence-free initialisation via potentialFoam.
- y^+ in the design range of the chosen wall functions.

What to be careful about.

- No curvature correction and no separation fix relative to HELYX. Expect faster vortex diffusion and possibly different separation onset.
- Relaxed mesh quality limits leave genuinely poor cells in the mesh (minTetQuality disabled). Accepted in exchange for layer coverage.

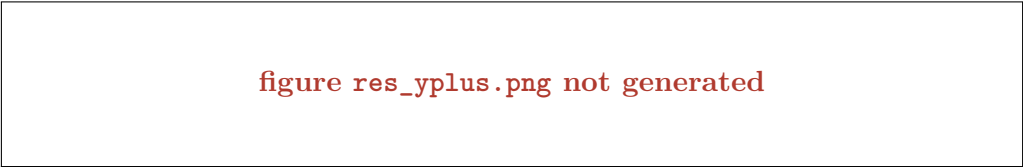


figure res_yplus.png not generated

Figure 20: y^+ on the wall patches. The layer stack targets $y^+ \approx 2.5$; the all- y^+ wall functions stay valid well beyond that, but values far above 30 mean the near-wall solution should not be trusted.

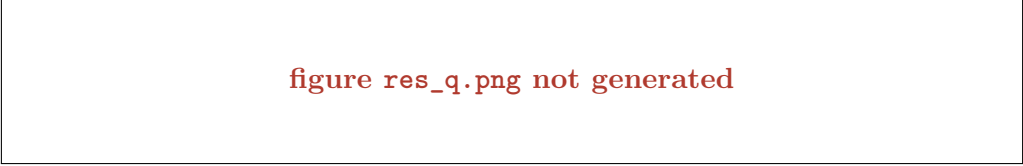


figure res_q.png not generated

Figure 21: Q-criterion isosurface, coloured by streamwise vorticity — the endplate and wheel vortex systems.

- Three patches without prism layers, see section 5.
- Two duplicate triangles in the wing STL.
- Steady RANS on a flow that may not be steady. At high angle of attack or low ride height, check the force spread before quoting a number.
- The two local refinement boxes that stand in for HELYX's per-region distance refinement are positioned for this geometry and do not follow the parameters. Re-check them before extending the sweep.